SAP Databricks

Generated on: 2026-04-15 16:42:37 GMT+0000

SAP Business Data Cloud | SHIP

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Original content: https://help.sap.com/docs/SAP_BUSINESS_DATA_CLOUD/3708ee482fc441ef8bb91711b1629109?locale=en-US&state=PRODUCTION&version=SHIP

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Introducing SAP Databricks

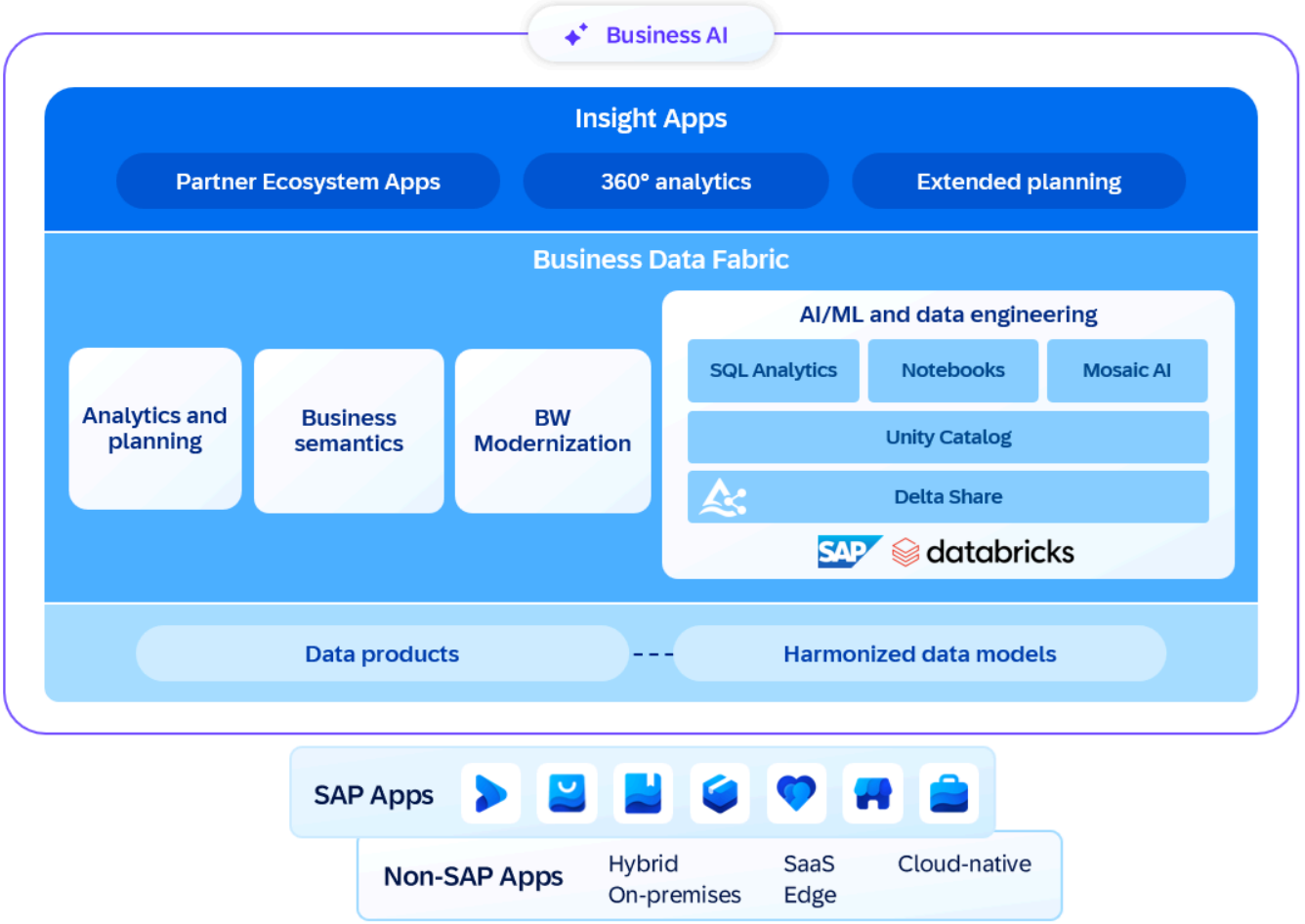
SAP Databricks is a special edition of Databricks available as an application in SAP Business Data Cloud. SAP Databricks provides Databricks' advanced AI and analytical capabilities in an environment that is fully integrated with your SAP landscape.

Key Benefits

- Use pre-configured Delta Sharing, access any data product in SAP Business Data Cloud from your SAP Databricks workspace without having to replicate them, eliminating the risk of data inconsistency.
- Use SAP Databricks to combine your SAP data with your external data, perform analysis, and publish data products from SAP back into SAP Business Data Cloud.
- Instantly provision SAP Databricks from SAP for Me with automatically integrated single sign-on and security, and manage your SAP Databricks account.
- SAP Databricks is a serverless offering, and compute and storage are fully managed. No additional cloud resources are needed.

Key Use Cases

- **Automated forecasting**: Build time-series forecasts with the best algorithms and hyperparameters automatically selected.
- **Fine-tuning an LLM**: Customize foundation models using your data to optimize performance for your specific applications.
- **Exploratory data analysis**: Use notebooks to collaborate in a multi-language work surface, allowing you to perform data science, visualizations, feature engineering, and ML model development.
- **Business Intelligence**: Use the SQL editor to run optimized queries and create visualizations for BI reporting.



SAP Databricks Security

This topic provides an overview of security features and capabilities available in SAP Databricks to secure your environment.

For further overview of Databricks security policies, see Networking and security in Databricks documentation.

Cloud Network and Communication Security

SAP Databricks supports the following networking and security features:

- IP access lists to restrict access to your Databricks account and workspaces based on a user's IP address.

  See IP Allowlisting below.

- Enable access from Databricks to other sources on your network using a network connectivity configuration (NCC).

  See Networking and security in Databricks documentation.

- Manage outbound network connectivity with serverless egress controls.

  See Connect to external data sources in Databricks documentation.

IP Allowlisting

By default, users can connect to SAP Databricks from any computer or IP address. IP access lists enable you to restrict access to your Databricks account and workspaces based on a user's IP address. See **Manage IP access lists** in Databricks documentation.

> **Caution**
> To avoid breaking the connection between SAP Business Data Cloud and SAP Databricks, check the list of SAP network access gateways before enforcing any IP ACLs (access control lists). See Domains and IP Ranges.

User Management

Authentication

Authentication to the SAP Databricks account is managed through SAP Cloud Identity Services. After your SAP Databricks account is provisioned, users are provisioned with SAP Cloud Identity Services, Identity Provisioning to SAP Databricks using SCIM. We recommend using your Identity Provisioning as the single source of truth for all users across your SAP Databricks account, see Identity Authentication. Additionally, account admins can directly add users who weren't synced through SAP Cloud Identity Services, see Managing SAP Databricks Users and Roles.

Access to workspace and access-control policies in SAP Databricks is managed via account-level groups. Groups can be created and managed in both your organization's SAP Cloud Identity Services environment and in SAP Databricks, and are synchronized bidirectionally between these systems, see Synchronizing Users and Groups with the SAP Databricks Account. You can add your managed groups to Databricks account groups.

User roles in SAP Databricks are defined and assigned via groups membership. This means that you can assign a specific role to a set of users by including them into the respective group associated with that role.

User Authorization

Authorization to data in SAP Databricks is governed through privileges to securable objects. Securable objects are objects in the Unity Catalog metastore on which privileges can be granted to a principal (user, service principal, or group). Permissions are managed and assigned in SAP Databricks, see Manage privileges in Unity Catalog in Databricks documentation.

SAP Databricks users must have the **USE PROVIDER** privilege to have access to data products in SAP Business Data Cloud.

Data Encryption

When you make data products available for consumption in SAP Databricks, a derived data product is created in the Unity Catalog of SAP Databricks. The data is stored inside a Unity-governed storage, and is protected by Unity governance. To store this data, SAP Databricks leverages HANA Cloud, Data Lake storage embedded in the product. The storage is SAP-managed, and is not customer-configurable. For details, see SAP HANA Cloud, Data Lake.

Data is in motion between SAP Business Data Cloud and SAP Databricks, is protected by secure mTLS-based transport encryption, which enhances the security of the TLS protocol by implementing two-way authentication and encryption.

SAP Databricks also provides compute storage for temporary data or runtime state, such as your notebook results and your queries. This data is also protected by a customer-specific encryption key that is managed and owned and operated by SAP.

Backup and Recovery

SAP Databricks automatically backs up your data and offers the ability to restore backed up data.

Backup

When data in SAP Databricks is overwritten or deleted, it is automatically retained in a snapshot. The retained data is held for a fixed period of 14 days and adds to the overall storage size of SAP Databricks. Snapshots are not visible when listing SAP Databricks contents.

Recovery

If you have data that you need to restore, please submit a case to SAP Support.

Currently SAP does not provide backups for compute storage. If a workspace hosting your notebooks or source code file is deleted, this data will be lost. SAP recommends that you use connected Git repositories for backing up notebooks and other critical data.

SAP Databricks Metering

SAP Databricks metering provides information on cloud subscription usage for SAP Public cloud products. Use it to understand and manage your subscriptions effectively, and to calculate the number of capacity units required.

To calculate the number of capacity units required for each SAP BDC capacity service, see SAP Business Data Cloud Capacity Unit Estimator.

To learn about your subscriptions across your SAP Public Cloud products in the **Public Cloud Consumption** card, see Public Cloud Consumption Card.

For information on cloud subscription usage, go to the page SAP Cloud Subscription Usage and search for **Capacity Units - SAP Business Data Cloud** (CY475). Choose the link **SAP Business Data Cloud**. In the PDF document that opens, search for **Capacity Units – SAP Databricks** (CY477).

For SAP Databricks capacity services, go to the page Cloud Services. Use the search bar to find the card **SAP Business Data Cloud Supplement**. Choose a card in one of the available languages. In that document, search for **Table 4: SAP Databricks Capacity Services**.

Logging In to SAP Databricks and Onboarding Users

Once the SAP Databricks tenant has been provisioned in the SAP for Me, and the user assigned the Owner role has activated the SAP Databricks account, this user can log in to the tenant via the main URL that provides access to the SAP Databricks selected workspace.

To onboard additional users to SAP Databricks, the user assigned the Owner role needs to share this link with the users targeted to have access to the SAP Databricks account. This can be done via various platforms and methods, depending on the needs and communication preferences of your organization.

For further actions, see Set up your SAP Databricks Account in Databricks documentation.

Managing SAP Databricks Users and Roles

Create, assign, and synchronize users and groups, through SAP Cloud Identity Services administrative console, allowing efficient administration of access and privileges within SAP Databricks.

1. **Registering Users**

   Create new users either manually, one by one, or a set of users by importing a CSV file.

2. **Assigning Groups and Roles**

   Define and assign roles via groups membership. Assign a specific role to a set of users by including them into the respective group associated with that role.

3. **Provisioning Identities**

Integrate SAP Databricks with SAP Cloud Identity Services securely to automate users and groups provisioning across systems.

4. **Synchronizing Users and Groups with the SAP Databricks Account**

   Perform the Resync Provisioning Job to synchronize users and groups between SAP Cloud Identity Services and SAP Databricks account. This ensures accurate user and group management.

5. **Defining New Roles**

   Assign relevant permissions and synchronize users and groups with SAP Databricks. Grant permissions in the Databricks environment and use it to manage SAP data products consumption.

Registering Users

Create new users either manually, one by one, or a set of users by importing a CSV file.

Context

Check the relevant SAP Cloud Identity Services documentation:

- Create a new user.
- Import users from CSV file.

To delete or deactivate users:

- Delete Users.
- Deactivate Users.

Task overview: Managing SAP Databricks Users and Roles

Next task: Assigning Groups and Roles

Assigning Groups and Roles

Define and assign roles via groups membership. Assign a specific role to a set of users by including them into the respective group associated with that role.

Context

You can create one or more groups in the SAP Cloud Identity Services tenant associated with your SAP Databricks tenant.

Example

- `databricks__account_admins`: Members of this group are **Databricks account administrators.**
- `databricks__workspace_default_users`: Members of this group are **users** of the `default` Databricks workspace.
- `databricks__workspace_default_admins`: Members of this group are **administrators** of the `default` SAP Databricks workspace. Check Databricks documentation on What are workspace admins? for more information.

- `databricks__data_product_publishers`: Members of this group are allowed to publish data products from SAP Databricks for consumption in SAP Business Data Cloud systems.

To assign users to groups, and grant them the respective role, see:

- Add Users to a Group
- Assign Groups to a User

To remove users from groups, and revoke their roles/privileges, see:

- Remove Users from a Group
- Unassign Groups from a User

Task overview: Managing SAP Databricks Users and Roles

Previous task: Registering Users

Next: Provisioning Identities

Provisioning Identities

Integrate SAP Databricks with SAP Cloud Identity Services securely to automate users and groups provisioning across systems.

In your SAP Databricks environment, you have to create a service principal, which acts as a dedicated, non-human identity that SAP Cloud Identity Services can use to access Databricks resources via APIs.

After creating the service principal, you'll prepare configuration JSON files that define how SAP Cloud Identity Services connects to Databricks for both reading and writing user and group data. These files map attributes, control transformations, and set up bidirectional identity synchronization between your local directory and Databricks.

In the SAP Cloud Identity Services environment, you have to upload these files as source systems, completing the link so identities can be provisioned and managed across platforms.

For information about provisioning and deprovisioning of users and groups in the SAP Cloud Identity Services, see Provisioning Systems.

1. **Creating Service Principals**

   Create service principals to give automated tools and scripts API-only access to SAP Databricks resources, providing greater security than using users accounts.

2. **Creating Provisioning Systems Configuration Files**

   Create JSON files that describe the configuration of provisioning systems to allow SAP Cloud Identity Services to manage and synchronize users and groups between SAP Databricks and the local identity directory of your SAP Cloud Identity Services tenant.

3. **Adding Source Systems**

   Add the source files to establish the connection between SAP Cloud Identity Services and SAP Databricks. This is required to enable secure synchronization of user and group data across both systems.

4. **Adding Target Systems**

   Add the target files to establish the connection between SAP Cloud Identity Services and SAP Databricks. This is required to enable secure synchronization of user and group data across both systems.

Parent topic: Managing SAP Databricks Users and Roles

Previous task: Assigning Groups and Roles

Next task: Synchronizing Users and Groups with the SAP Databricks Account

Creating Service Principals

Create service principals to give automated tools and scripts API-only access to SAP Databricks resources, providing greater security than using users accounts.

Context

You can grant SAP Databricks users and groups permissions to use a service principal. This allows users to run jobs as the service principal, instead of as their identity, which prevents jobs from failing if a user leaves your organization or a group is modified.

Follow the procedure below to create a service principal for SAP Cloud Identity Services in your SAP Databricks account.

Procedure

1. Go to **User Management** > **Service principals** > **Add service principal**.
2. In the pop-up screen, enter `SAP IPS` and choose **Add service principal**.
3. In the next screen, choose the **Credentials & secrets** tab and choose the **Generate secret** button.
4. In the pop-up screen, enter the lifetime and choose **Generate**.

   The system generates an OAuth secret.
5. Copy the **Secret** and the **Client ID** to a secure location, like a notepad, and then choose **Done**.

   > **Note**
   > You will not be able to view them again.

6. Choose the **Roles** tab and enable the **Account admin** toggle.

Task overview: Provisioning Identities

Next task: Creating Provisioning Systems Configuration Files

Creating Provisioning Systems Configuration Files

Create JSON files that describe the configuration of provisioning systems to allow SAP Cloud Identity Services to manage and synchronize users and groups between SAP Databricks and the local identity directory of your SAP Cloud Identity Services tenant.

Prerequisites

You have created a secret and a client ID. See Creating Service Principals.

Context

To prepare for the next tasks, create four files using a text editor.

Procedure

Create File #1

1. Create a new file in your default, preinstalled, or preferred text editor.
2. Expand the **Code Sample** below. Then, copy and paste it in your text editor:

Source Code

```
{
    "id": "4aceb7db-8e75-40b5-952b-ce0c000dd6b3",
    "connectorTypeString": "DATABRICKS",
    "accessMode": "READ",
    "destinationName": "",
    "gitAllowedExpressions": [],
    "gitDisallowedExpressions": [],
    "emailSubscribers": [],
    "name": "Databricks Source - {{DATABRICKS_ACCOUNT_ID}}",
    "tenantId": "as9v97bjl",
    "state": "ENABLED",
    "systemManagementType": "CUSTOMER_MANAGED",
    "properties": {
        "User": {{null}},
        "OAuth2TokenScope": "all-apis",
        "OAuth2TokenServiceURL": "https://accounts.{{cloud}}.databricks.com/oidc/accounts/{
        "URL": "https://accounts.{{cloud}}.databricks.com/api/2.0/accounts/{{DATABRICKS_ACC
        "Authentication": "BasicAuthentication",
        "Type": "HTTP",
        "ProxyType": "Internet",
        "ips.trace.failed.entity.content": "false"
    },
    "encryptedProperties": {
        "Password": {{null}}
    },
    "automaticOutboundCertificateRenew": false,
    "transformationChanged": true,
    "readTransformationChanged": false,
    "writeTransformationChanged": false,
    "fromImport": false,
    "transformation": {
        "user": {
            "ignore": false,
            "mappings": [
                {
                    "sourcePath": "$",
                    "targetPath": "$"
                },
                {
                    "sourcePath": "$.id",
                    "targetVariable": "entityIdSourceSystem"
                },
                {
                    "sourcePath": "$.userName",
                    "correlationAttribute": true
```

```
                    },
                    {
                        "sourcePath": "$.emails[?(@.primary== true)].value",
                        "optional": true,
                        "correlationAttribute": true
                    },
                    {
                        "targetPath": "$.id",
                        "type": "remove"
                    },
                    {
                        "targetPath": "$.meta",
                        "type": "remove"
                    }
                ]
            },
            "group": {
                "ignore": false,
                "condition": "$.displayName != 'account users'",
                "mappings": [
                    {
                        "sourcePath": "$",
                        "targetPath": "$"
                    },
                    {
                        "sourcePath": "$.id",
                        "targetVariable": "entityIdSourceSystem"
                    },
                    {
                        "targetPath": "$.id",
                        "type": "remove"
                    },
                    {
                        "targetPath": "$.meta",
                        "type": "remove"
                    },
                    {
                        "sourcePath": "$.displayName",
                        "targetPath": "$.displayName",
                        "functions": [
                            {
                                "function": "concatString",
                                "prefix": "Databricks_{{DATABRICKS_ACCOUNT_ID}}:"
                            }
                        ]
                    }
                ]
            }
        },
        "gitFetchAllowed": false
    }
```

3. In the code you just pasted, make the following changes:

- Replace all the 4 entries above labeled as **{{DATABRICKS_ACCOUNT_ID}}** with your real SAP Databricks account ID.

  ### Note

  You can find your ID in the SAP Databricks URL. Copy the part that is after **account_id=**.

  ### Example

  Your SAP Databricks URL is https://accounts.cloud.databricks.com/?account_id=85ac9fa7-2f06-43a7-8c47-7329e91cd90d.

  You must copy **85ac9fa7-2f06-43a7-8c47-7329e91cd90d**, which is your account ID.

- In the **"User"**, replace **{{null}}** with the client ID you created.
- In the **"Password"**, replace **{{null}}** with the secret you created.
- In the **"OAuth2TokenServiceURL"** and the **"URL"**, replace **{{cloud}}** with **azure** or **gcp** if you use Azure or Google Cloud Platform (GCP) as your hyperscaler. In case you use a different hyperscaler, simply remove the brackets of **{{cloud}}**.

  ### Example

  - Azure: `https://accounts.azure.databricks.com/...`
  - GCP: `https://accounts.gcp.databricks.com/...`
  - Others: `https://accounts.cloud.databricks.com/...`

4. Save this file in JSON format. This is your **Databricks Source** file. You can name it like that.

   You created your first file.

Create File #2

5. Create a new file in your text editor.
6. Expand the Code Sample below. Then, copy and paste it in your text editor:

Source Code

```
{
    "id": "a6a7b1c8-90f7-4296-9fd0-fc97cd996420",
    "connectorTypeString": "DATABRICKS",
    "accessMode": "WRITE",
    "destinationName": "",
    "relatedSystems": [
        "Local Identity Directory Source - {{DATABRICKS_ACCOUNT_ID}}"
    ],
    "gitAllowedExpressions": [],
    "gitDisallowedExpressions": [],
    "emailSubscribers": [],
    "name": "Databricks Target - {{DATABRICKS_ACCOUNT_ID}}",
    "tenantId": "as9v97bjl",
    "state": "ENABLED",
    "systemManagementType": "CUSTOMER_MANAGED",
    "properties": {
        "User": {{null}},
        "OAuth2TokenScope": "all-apis",
```

```
            "OAuth2TokenServiceURL": "https://accounts.{{cloud}}.databricks.com/oidc/accounts/{
            "URL": "https://accounts.{{cloud}}.databricks.com/api/2.0/accounts/{{DATABRICKS_ACC
            "Authentication": "BasicAuthentication",
            "scim.support.patch.operation": "true",
            "scim.user.unique.attribute": "userName",
            "scim.group.unique.attribute": "displayName",
            "Type": "HTTP",
            "ProxyType": "Internet",
            "ips.trace.failed.entity.content": "false"
        },
        "encryptedProperties": {
            "Password": {{null}}
        },
        "automaticOutboundCertificateRenew": false,
        "transformationChanged": true,
        "readTransformationChanged": false,
        "writeTransformationChanged": false,
        "fromImport": false,
        "transformation": {
            "user": {
                "ignore": false,
                "mappings": [
                    {
                        "sourcePath": "$.active",
                        "targetPath": "$.active"
                    },
                    {
                        "condition": "($.displayName != 'None') && ($.displayName EMPTY false)'
                        "sourcePath": "$.displayName",
                        "targetPath": "$.displayName"
                    },
                    {
                        "sourcePath": "$.emails",
                        "preserveArrayWithSingleElement": true,
                        "targetPath": "$.emails"
                    },
                    {
                        "condition": "$.emails[0].length() > 0",
                        "constant": true,
                        "targetPath": "$.emails[0].primary"
                    },
                    {
                        "targetPath": "$.id",
                        "type": "remove"
                    },
                    {
                        "sourceVariable": "entityIdTargetSystem",
                        "targetPath": "$.id"
                    },
                    {
                        "sourcePath": "$.name",
                        "targetPath": "$.name"
                    },
                    {
```

```
                "constant": [
                    "urn:ietf:params:scim:schemas:core:2.0:User",
                    "urn:ietf:params:scim:schemas:extension:enterprise:2.0:User"
                ],
                "targetPath": "$.schemas"
            },
            {
                "sourcePath": "$.emails[0].value",
                "optional": false,
                "targetPath": "$.userName"
            },
            {
                "sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
                "optional": true,
                "targetPath": "$.externalId"
            },
            {
                "targetPath": "$['urn:ietf:params:scim:schemas:extension:enterprise:2.0
                "type": "remove"
            }
        ]
    },
    "group": {
        "ignore": false,
        "mappings": [
            {
                "sourcePath": "$.displayName",
                "targetPath": "$.displayName",
                "functions": [
                    {
                        "function": "replaceString",
                        "target": "Databricks_{{DATABRICKS_ACCOUNT_ID}}:",
                        "replacement": ""
                    }
                ]
            },
            {
                "targetPath": "$.id",
                "type": "remove"
            },
            {
                "sourceVariable": "entityIdTargetSystem",
                "targetPath": "$.id"
            },
            {
                "condition": "$.members EMPTY true",
                "targetPath": "$.members",
                "constant": []
            },
            {
                "sourcePath": "$.members[*].value",
                "preserveArrayWithSingleElement": true,
                "optional": true,
                "targetPath": "$.members[?(@.value)]",
```

```
                        "functions": [
                            {
                                "function": "resolveEntityIds",
                                "entityType": "user"
                            }
                        ]
                    },
                    {
                        "sourcePath": "$.members[*].value",
                        "preserveArrayWithSingleElement": true,
                        "optional": true,
                        "targetPath": "$.members[?(@.value)]",
                        "functions": [
                            {
                                "function": "resolveEntityIds",
                                "entityType": "group"
                            }
                        ]
                    },
                    {
                        "constant": [
                            "urn:ietf:params:scim:schemas:core:2.0:Group"
                        ],
                        "targetPath": "$.schemas"
                    },
                    {
                        "constant": [],
                        "targetPath": "$.roles",
                        "scope": "createEntity"
                    }
                ]
            }
        },
        "gitFetchAllowed": false
    }
```

7. In the code you just pasted, make the following changes:
    - Replace all the 4 entries above labeled as **{{DATABRICKS_ACCOUNT_ID}}** with your real SAP Databricks account ID.

      ### ℹ Note

      You can find your ID in the SAP Databricks URL. Copy the part that is after **account_id=**.

      ### Example

      Your SAP Databricks URL is https://accounts.cloud.databricks.com/?account_id=85ac9fa7-2f06-43a7-8c47-7329e91cd90d.

      You must copy **85ac9fa7-2f06-43a7-8c47-7329e91cd90d**, which is your account ID.

    - In the **"User"**, replace **{{null}}** with the client ID you created.
    - In the **"Password"**, replace **{{null}}** with the secret you created.

- In the **"OAuth2TokenServiceURL"** and the **"URL"**, replace **{{cloud}}** with **azure** or **gcp** if you use Azure or Google Cloud Platform (GCP) as your hyperscaler. In case you use a different hyperscaler, simply remove the brackets of **{{cloud}}**.

Example

- Azure: `https://accounts.azure.databricks.com/...`
- GCP: `https://accounts.gcp.databricks.com/...`
- Others: `https://accounts.cloud.databricks.com/...`

8. Save this file in JSON format. This is your **Databricks Target** file. You can name it like that.

   You created your second file.

Create File #3

9. Create a new file in your text editor.
10. Expand the Code Sample below. Then, copy and paste it in your text editor:

Source Code

```
{
    "id": "d8c35da9-4d64-4d41-b0d7-83f7ec7bd4f1",
    "connectorTypeString": "LOCAL_IDENTITY_DIRECTORY_CC",
    "accessMode": "READ",
    "destinationName": "",
    "gitAllowedExpressions": [],
    "gitDisallowedExpressions": [],
    "emailSubscribers": [],
    "name": "Local Identity Directory Source - {{DATABRICKS_ACCOUNT_ID}}",
    "tenantId": "as9v97bjl",
    "state": "ENABLED",
    "systemManagementType": "CUSTOMER_MANAGED",
    "properties": {
        "idds.group.filter": "displayName sw \"Databricks_{{DATABRICKS_ACCOUNT_ID}}:\"",
        "idds.user.filter": "groups.display sw \"Databricks_{{DATABRICKS_ACCOUNT_ID}}:\"",
        "ips.trace.failed.entity.content": "false"
    },
    "encryptedProperties": {},
    "automaticOutboundCertificateRenew": false,
    "transformationChanged": false,
    "readTransformationChanged": false,
    "writeTransformationChanged": false,
    "fromImport": false,
    "transformation": {
        "user": {
            "mappings": [
                {
                    "sourcePath": "$.id",
                    "targetPath": "$['urn:sap:cloud:scim:schemas:extension:custom:2.0:User'
                    "targetVariable": "entityIdSourceSystem"
                },
                {
                    "sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
```

```
            "targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
        },
        {
            "sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
            "targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
        },
        {
            "sourcePath": "$.schemas",
            "preserveArrayWithSingleElement": true,
            "targetPath": "$.schemas"
        },
        {
            "sourcePath": "$.userName",
            "optional": true,
            "targetPath": "$.userName",
            "correlationAttribute": true
        },
        {
            "sourcePath": "$.name.givenName",
            "optional": true,
            "targetPath": "$.name.givenName"
        },
        {
            "sourcePath": "$.name.middleName",
            "optional": true,
            "targetPath": "$.name.middleName"
        },
        {
            "sourcePath": "$.name.familyName",
            "optional": true,
            "targetPath": "$.name.familyName"
        },
        {
            "sourcePath": "$.name.honorificPrefix",
            "optional": true,
            "targetPath": "$.name.honorificPrefix"
        },
        {
            "sourcePath": "$.emails",
            "preserveArrayWithSingleElement": true,
            "targetPath": "$.emails"
        },
        {
            "sourcePath": "$.emails[?(@.primary== true)].value",
            "correlationAttribute": true
        },
        {
            "sourcePath": "$.active",
            "targetPath": "$.active"
        },
        {
            "sourcePath": "$.userType",
            "optional": true,
            "targetPath": "$.userType"
```

```
                    },
                    {
                        "sourcePath": "$.addresses",
                        "preserveArrayWithSingleElement": true,
                        "optional": true,
                        "targetPath": "$.addresses"
                    },
                    {
                        "sourcePath": "$.locale",
                        "optional": true,
                        "targetPath": "$.locale"
                    },
                    {
                        "sourcePath": "$.phoneNumbers",
                        "preserveArrayWithSingleElement": true,
                        "optional": true,
                        "targetPath": "$.phoneNumbers"
                    },
                    {
                        "sourcePath": "$.timezone",
                        "optional": true,
                        "targetPath": "$.timezone"
                    },
                    {
                        "sourcePath": "$.displayName",
                        "optional": true,
                        "targetPath": "$.displayName"
                    },
                    {
                        "sourcePath": "$.groups",
                        "preserveArrayWithSingleElement": true,
                        "optional": true,
                        "targetPath": "$.groups"
                    },
                    {
                        "condition": "$.displayName EMPTY true",
                        "targetPath": "$.displayName",
                        "type": "remove"
                    },
                    {
                        "sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
                        "optional": true,
                        "targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
                    },
                    {
                        "sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
                        "optional": true,
                        "targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
                    },
                    {
                        "sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
                        "optional": true,
                        "targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
                    },
```

```
                {
                    "sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
                    "optional": true,
                    "targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:User']
                },
                {
                    "sourcePath": "$['urn:ietf:params:scim:schemas:extension:enterprise:2.0
                    "optional": true,
                    "targetPath": "$['urn:ietf:params:scim:schemas:extension:enterprise:2.0
                },
                {
                    "sourcePath": "$['urn:ietf:params:scim:schemas:extension:enterprise:2.0
                    "optional": true,
                    "targetPath": "$['urn:ietf:params:scim:schemas:extension:enterprise:2.0
                },
                {
                    "sourcePath": "$['urn:ietf:params:scim:schemas:extension:enterprise:2.0
                    "optional": true,
                    "targetPath": "$['urn:ietf:params:scim:schemas:extension:enterprise:2.0
                },
                {
                    "sourcePath": "$['urn:ietf:params:scim:schemas:extension:enterprise:2.0
                    "optional": true,
                    "targetPath": "$['urn:ietf:params:scim:schemas:extension:enterprise:2.0
                },
                {
                    "sourcePath": "$['urn:ietf:params:scim:schemas:extension:enterprise:2.0
                    "optional": true,
                    "targetPath": "$['urn:ietf:params:scim:schemas:extension:enterprise:2.0
                },
                {
                    "sourcePath": "$['urn:ietf:params:scim:schemas:extension:enterprise:2.0
                    "optional": true,
                    "targetPath": "$['urn:ietf:params:scim:schemas:extension:enterprise:2.0
                },
                {
                    "sourcePath": "$['urn:ietf:params:scim:schemas:extension:enterprise:2.0
                    "optional": true,
                    "targetPath": "$['urn:ietf:params:scim:schemas:extension:enterprise:2.0
                },
                {
                    "sourcePath": "$['urn:ietf:params:scim:schemas:extension:enterprise:2.0
                    "optional": true,
                    "targetPath": "$['urn:ietf:params:scim:schemas:extension:enterprise:2.0
                },
                {
                    "sourcePath": "$['urn:sap:cloud:scim:schemas:extension:custom:2.0:User'
                    "optional": true,
                    "targetPath": "$['urn:sap:cloud:scim:schemas:extension:custom:2.0:User'
                },
                {
                    "sourcePath": "$.company",
                    "optional": true,
                    "targetPath": "$['urn:ietf:params:scim:schemas:extension:enterprise:2.0
                }
            ]
        },
        "group": {
            "mappings": [
```

```
                        {
                            "sourcePath": "$.id",
                            "targetVariable": "entityIdSourceSystem"
                        },
                        {
                            "sourcePath": "$['urn:sap:cloud:scim:schemas:extension:custom:2.0:Grou
                            "targetPath": "$['urn:sap:cloud:scim:schemas:extension:custom:2.0:Grou
                        },
                        {
                            "sourcePath": "$.displayName",
                            "targetPath": "$.displayName"
                        },
                        {
                            "sourcePath": "$.members",
                            "preserveArrayWithSingleElement": true,
                            "optional": true,
                            "targetPath": "$.members"
                        },
                        {
                            "sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group'
                            "optional": true,
                            "targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group'
                        },
                        {
                            "sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group'
                            "optional": true,
                            "targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group'
                        },
                        {
                            "sourcePath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group'
                            "optional": true,
                            "targetPath": "$['urn:ietf:params:scim:schemas:extension:sap:2.0:Group'
                        }
                    ]
                }
            },
            "gitFetchAllowed": false
        }
```

11. In the code you just pasted, replace all the 3 {{DATABRICKS_ACCOUNT_ID}} entries with your real SAP Databricks account ID.
12. Save this file in JSON format. This is your **Local Source** file. You can name it like that.

    You created your third file.

Create File #4

13. Create a new file in your text editor.
14. Expand the **Code Sample** below. Then, copy and paste it in your text editor:

Source Code

```
{
    "id": "829dcc82-0d63-4aa9-977f-8ded1b19abd7",
```

```
            "connectorTypeString": "LOCAL_IDENTITY_DIRECTORY_CC",
            "accessMode": "WRITE",
            "destinationName": "",
            "relatedSystems": [
                "Databricks Source - {{DATABRICKS_ACCOUNT_ID}}"
            ],
            "gitAllowedExpressions": [],
            "gitDisallowedExpressions": [],
            "emailSubscribers": [],
            "name": "Local Identity Directory Target - {{DATABRICKS_ACCOUNT_ID}}",
            "tenantId": "as9v97bjl",
            "state": "ENABLED",
            "systemManagementType": "CUSTOMER_MANAGED",
            "properties": {
                "idds.group.unique.attribute": "displayName",
                "idds.user.unique.attribute": "emails[0].value",
                "idds.support.patch.operation": "true",
                "ips.trace.failed.entity.content": "false"
            },
            "encryptedProperties": {},
            "automaticOutboundCertificateRenew": false,
            "transformationChanged": true,
            "readTransformationChanged": false,
            "writeTransformationChanged": false,
            "fromImport": false,
            "transformation": {
                "user": {
                    "ignore": false,
                    "condition": "($.emails EMPTY false) && ($.userName EMPTY false) && isValidEmai
                    "skipOperations": [
                        "create",
                        "delete"
                    ],
                    "mappings": [
                        {
                            "sourcePath": "$.userName",
                            "targetPath": "$.userName"
                        },
                        {
                            "sourcePath": "$.userName",
                            "targetPath": "$.emails[0].value"
                        }
                    ]
                },
                "group": {
                    "ignore": false,
                    "mappings": [
                        {
                            "sourceVariable": "entityIdTargetSystem",
                            "targetPath": "$.id"
                        },
                        {
                            "constant": [
                                "urn:ietf:params:scim:schemas:core:2.0:Group",
```

```
                        "urn:sap:cloud:scim:schemas:extension:custom:2.0:Group"
                    ],
                    "targetPath": "$.schemas"
                },
                {
                    "sourcePath": "$.displayName",
                    "targetPath": "$.displayName"
                },
                {
                    "sourcePath": "$.members[*].value",
                    "preserveArrayWithSingleElement": true,
                    "optional": true,
                    "targetPath": "$.members[?(@.value)]",
                    "functions": [
                        {
                            "function": "resolveEntityIds",
                            "entityType": "user"
                        }
                    ]
                },
                {
                    "sourcePath": "$.members[*].value",
                    "preserveArrayWithSingleElement": true,
                    "optional": true,
                    "targetPath": "$.members[?(@.value)]",
                    "functions": [
                        {
                            "function": "resolveEntityIds",
                            "entityType": "group"
                        }
                    ]
                },
                {
                    "sourcePath": "$.displayName",
                    "targetPath": "$['urn:sap:cloud:scim:schemas:extension:custom:2.0:Grou
                    "scope": "createEntity"
                },
                {
                    "sourcePath": "$['urn:sap:cloud:scim:schemas:extension:custom:2.0:Grou
                    "optional": true,
                    "targetPath": "$['urn:sap:cloud:scim:schemas:extension:custom:2.0:Grou
                    "scope": "createEntity"
                },
                {
                    "sourcePath": "$['urn:sap:cloud:scim:schemas:extension:custom:2.0:Grou
                    "optional": true,
                    "targetPath": "$['urn:sap:cloud:scim:schemas:extension:custom:2.0:Grou
                }
            ]
        }
    },
    "gitFetchAllowed": false
}
```

15. In the code you just pasted, replace the 2 {{DATABRICKS_ACCOUNT_ID}} entries with your real SAP Databricks account ID.
16. Save this file in JSON format. This is your **Local Target** file. You can name it like that.

    You created your fourth and last file.

Task overview: Provisioning Identities

Previous task: Creating Service Principals

Next task: Adding Source Systems

Adding Source Systems

Add the source files to establish the connection between SAP Cloud Identity Services and SAP Databricks. This is required to enable secure synchronization of user and group data across both systems.

Prerequisites

You have created the provisioning systems configuration files. See Creating Provisioning Systems Configuration Files.

Procedure

1. In the SAP Cloud Identity Services environment, choose the **Identity Provisioning** tab and, from the dropdown menu, choose **Source Systems**.
2. From the sidebar menu, choose **Add** and then use the **Browse** button to import the **first** file you prepared previously. This would be your **Databricks Source** file.
3. Once you have uploaded the file, choose the **Properties** button. In the field **Password**, enter the secret and, in the field **User**, enter the client ID.
4. Choose **Save**.
5. From the sidebar menu, choose **Add** and then use the **Browse** button to import the **third** file you prepared previously. This would be your **Local Source** file.
6. Once you have uploaded the file, choose the **Properties** button. In the field **Password**, enter the secret and, in the field **User**, enter the client ID.
7. Choose **Save**.

Task overview: Provisioning Identities

Previous task: Creating Provisioning Systems Configuration Files

Next task: Adding Target Systems

Adding Target Systems

Add the target files to establish the connection between SAP Cloud Identity Services and SAP Databricks. This is required to enable secure synchronization of user and group data across both systems.

Prerequisites

You have created the provisioning systems configuration files. See Creating Provisioning Systems Configuration Files.

Procedure

1. In the SAP Cloud Identity Services environment, choose the **Identity Provisioning** tab and, from the dropdown menu, choose **Target Systems**.
2. From the sidebar menu, choose **Add** and then use the **Browse** button to import the **second** file you prepared previously. This would be your **Databricks Target** file.
3. Once you have uploaded the file, choose the **Properties** button. In the field **Password**, enter the secret and, in the field **User**, enter the client ID.
4. Choose **Save**.
5. From the sidebar menu, choose **Add** and then use the **Browse** button to import the **fourth** file you prepared previously. This would be your **Local Target** file.
6. Once you have uploaded the file, choose the **Properties** button. In the field **Password**, enter the secret and, in the field **User**, enter the client ID.
7. Choose **Save**.

Task overview: Provisioning Identities

Previous task: Adding Source Systems

Synchronizing Users and Groups with the SAP Databricks Account

Perform the Resync Provisioning Job to synchronize users and groups between SAP Cloud Identity Services and SAP Databricks account. This ensures accurate user and group management.

Context

After registering users and assigning them to the corresponding groups, you can perform the Resync Provisioning Job to synchronize and propagate the users and groups defined in the SAP Cloud Identity Services tenant to the SAP Databricks account. See Resync Provisioning Job.

Procedure

1. Go to **Identity Provisioning** > **Source Systems**.
2. Select the **Local IAS Source** system and go to **Jobs** > **Resync Job** > **Run Now**.

   Note

   To check the status of the synchronization job, go to **Identity Provisioning** > **Provisioning Logs**. Once the job run successfully, you can check the admin console of the Databricks account (under the **User Management** page) to check whether the new users were created and the group membership was assigned correctly.

Task overview: Managing SAP Databricks Users and Roles

Previous: Provisioning Identities

Next task: Defining New Roles

Defining New Roles

Assign relevant permissions and synchronize users and groups with SAP Databricks. Grant permissions in the Databricks environment and use it to manage SAP data products consumption.

Context

In addition to the pre-configured groups/roles, you can define new roles by creating new groups and granting them the relevant privileges in the SAP Databricks account. To do so, create a new role/group named **Data Product Admin** so that users are authorized to manage SAP data products consumption in SAP Databricks.

Procedure

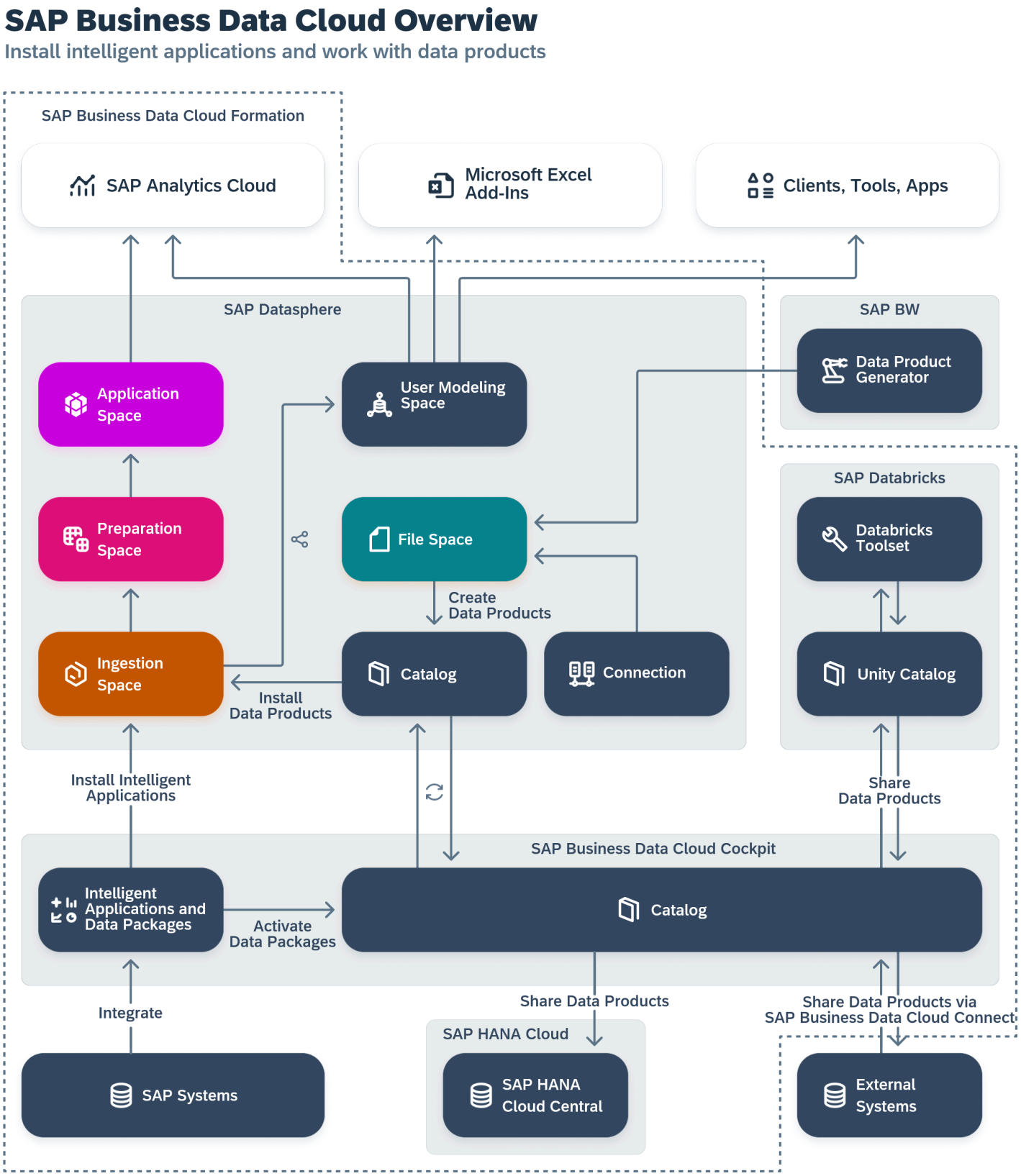
1. To create a new group in SAP Cloud Identity Services, see Create a New Group and:
   a. In the **Cloud Identity Services** admin console, go to **Users & Authorizations** > **Groups** > **Create**.
   b. Fill in the fields **Name** and **Display Name** of the group with the following pattern: `databricks__<role-name>` (example: `databricks__data_product_admins`, note the required `databricks__` prefix).
2. Assign this role to the relevant users by adding them as members of the group. See Assigning Groups and Roles.
3. Synchronize the users and groups from SAP Cloud Identity Services into SAP Databricks. See Synchronizing Users and Groups with the SAP Databricks Account.
4. Assign the relevant permissions to this group in the SAP Databricks environment. For the role `Data Product Admin` specifically, users must be granted with the following permissions on the **Unity Catalog Metastore**: Select the boxes **CREATE CATALOG**, **CREATE SHARE**, **SET SHARE PERMISSION**, **USE PROVIDER**, **USE RECIPIENT**, and **USE SHARE** then choose **Confirm**. For further details on **Unity Catalog** permissions, see Unity Catalog Privileges and Securable Objects and Manage Privileges in Unity Catalog .
   a. Access the **Workspace** UI and open the **Catalog Explorer** (in the sidebar, choose **Catalog**).
   b. At the top of the **Catalog Explorer** view, choose the **Gear** icon and select the **Metastore**.
   c. Select the **Permissions** tab and choose **Grant**.
   d. Type and select the **Principal** you want to use (`__data_product_admins`) and select the boxes **CREATE CATALOG**, **CREATE SHARE**, **SET SHARE PERMISSION**, **USE PROVIDER**, **USE RECIPIENT**, and **USE SHARE** then choose **Confirm**.

Task overview: Managing SAP Databricks Users and Roles

Previous task: Synchronizing Users and Groups with the SAP Databricks Account

Working with Data Products in SAP Databricks

Users with an appropriate SAP Databricks role can consume, query, process, and transform data products in SAP Databricks, including working with SQL and Python notebooks and using the data products for various data science and AI/ML projects.



To prepare for working with data products in SAP Databricks, a user with an administrator role must:

- Add the SAP Databricks tenant to the formation (see Creating SAP Business Data Cloud Formations).
- Activate one or more data packages (see Activate Data Packages).
- Provide access to the SAP Databricks tenant (see Manage Users and Roles).

Users with an appropriate SAP Databricks role can:

- Learn more about what SAP Databricks is and how it can help you reach your goals (see Introducing SAP Databricks).

- Access data products that are shared from the catalog (see Sharing Data Products to SAP Databricks).
- Install data products directly from the Unity Catalog (see Sharing SAP Business Data Cloud Data Products to SAP Databricks).
- Use the full SAP Databricks toolset to work with your data products (see Get started with Databricks in the *Databricks* documentation).
- Share data products to the SAP Datasphere Catalog (see Sharing Data Products from SAP Databricks to SAP Business Data Cloud).

Adding Privileges into SAP Databricks

You can manage and share data assets and notebooks efficiently in your SAP Databricks environment.

Context

After provisioning SAP Databricks, you must grant users permissions to manage data.

Procedure

1. Access your SAP Databricks Workspace environment. From the sidebar menu, choose **Catalog**.
2. Choose the **Manage** (Gear icon) button and, from the dropdown menu, select **Metastore**.
3. Choose the **Permissions** tab and then choose the **Grant** button.
4. In the **Principals** field, enter the user or a group of users.
5. Select the boxes **CREATE CATALOG**, **CREATE SHARE**, **SET SHARE PERMISSION**, **USE PROVIDER**, **USE RECIPIENT**, and **USE SHARE** then choose **Confirm**.

   You have added the necessary privileges.

Sharing SAP Business Data Cloud Data Products to SAP Databricks

To make data products available for consumption in SAP Databricks, you can either share data products from the SAP Business Data Cloud catalog to SAP Databricks, or in SAP Databricks you can create a catalog from the Delta Share associated with the desired data product.

Sharing Data Products from the SAP Business Data Cloud Catalog to SAP Databricks

Prerequisites

- You have added the necessary privileges. See Adding Privileges into SAP Databricks.

  To search for and share data products, you must have a global role that grants you the following privileges:

  - **Catalog Asset** (–R——- -) - To access the catalog and view objects in the **Assets** and **Data Products** collections.
  - **Cloud Data Product** (- - - -–S -) - To share data products to target systems.

  You can apply the **Catalog Administrator** and the **BD Viewer** roles together, for example, to grant these privileges (see Standard Roles Delivered with SAP Business Data Cloud and Privileges and Permissions in SAP Business Data Cloud).

- A user with an administrator role have activated in SAP Business Data Cloud cockpit the data packages in which these data products are included (see Activating Data Packages).

Context

In the central SAP Business Data Cloud catalog, you can select a data product to share to SAP Databricks.

Procedure

1. In the side navigation area, go to **Catalog & Marketplace** > **Search**.
2. On the catalog search page, use the filters or the search to find the data product you want and select it to open its details page. For more information, see Searching for Data Products and Assets in the Catalog.

   In the **Filter** panel, select the **Data Products** collection and the **Shareable** share status.
3. On the data product details page choose the tab **Overview** > **Details** to review the list of APIs.
4. For the row you want, select the **Share** button to open the **Manage Share Access** dialog.

   → **Tip**

   - If you're still viewing the search results in grid or table view, you can select **Share** for the data product.
   - If you're reviewing the API details page, you can select the **Share** button in the toolbar.

   In the **Overview** section, you can learn more about the data product by review its details and available objects.
5. Under **Target System**, do the following:

   a. If more than one target system is available, select the system. If only one target is available, it's automatically selected.

   b. Enter a name. This name will be the name of the delta share that appears in the target system

   c. You can share the catalog with up to five workspaces. Users in the selected workspaces can access the data product.
6. Select **Share**.

   A message appears letting you know that the share process has started. After the process finishes, a notification appears letting you know the result.

Results

Your data product is shared and its share status is updated and visible in the catalog search results. The name that you entered is the name that users will see in the target system, and it cannot be changed. If you need to change the name, you must delete the share setting for the data product and then share it to the target system again.

In SAP Databricks, users who have authorized access to your data product will be able to utilize the full SAP Databricks toolset to work with it (see Working with Data Products in SAP Databricks).

For information on consuming and processing the data products in SAP Databricks, see Creating and Working with Derived Data Products in SAP Databricks. For information on sharing these data products back to SAP Business Data Cloud, see Sharing Data Products from SAP Databricks to SAP Business Data Cloud.

Creating a SAP Databricks Delta Sharing Catalog from a Data Product

Check SAP Databricks documentation: Create a catalog from a share .

Prerequisites

The user must be a **Data Product Admin** and have the following privileges on the SAP Databricks Metastore: **USE PROVIDER**, **CREATE_CATALOG**. For more details on this role, see Managing SAP Databricks Users and Roles.

Procedure

1. Access the SAP Databricks workspace and open the Catalog Explorer.
2. At the top of the Catalog pane, choose the **Gear** icon and select **Delta Sharing**.
3. On the **Shared with me** tab, find and select the provider corresponding to the Data Product Provider system that is included into the formation, along with your SAP Databricks tenant.

   > **Note**
   > To correlate the providers with the systems in the formation, check the value in the **Comment** column.

4. On the **Shares** tab, find the share and choose **Create catalog**.
   a. Enter a name for the catalog.
   b. Choose **Create**.
5. After the catalog is created, it will be listed under the **Shared** section in the main view of the Catalog Explorer.
6. [OPTIONAL] To grant other users access to read data from this catalog:
   a. Select the catalog name in the Catalog Explorer under the **Shared** section.
   b. Access the **Permissions** tab and choose **Grant**.
   c. Type and select the desired Principal and mark the Privileges: **USE CATALOG**, **BROWSE**, **SELECT**.

For more information, see SAP Databricks documentation: Create a catalog from a share .

Consuming and Processing SAP Business Data Cloud Data Products in SAP Databricks

After a Delta Sharing catalog is created from a Data Product share, the data can be consumed, queried, processed, transformed by authorized users as a regular catalog in the Unity Catalog. In the next sub-sections, you can find some examples of how to query data using notebooks in SQL and Python. Check SAP Databricks documentation for extra information::

- Query data
- Transform data
- Use Delta Lake change data feed

Permissions

To read the data from the SAP Data Product, the user must have the following privileges on the Delta Sharing catalog associated with the SAP Data Product (granted by the catalog owner): **USE CATALOG**, **BROWSE**, **SELECT**.

Querying Data in a Python Notebook

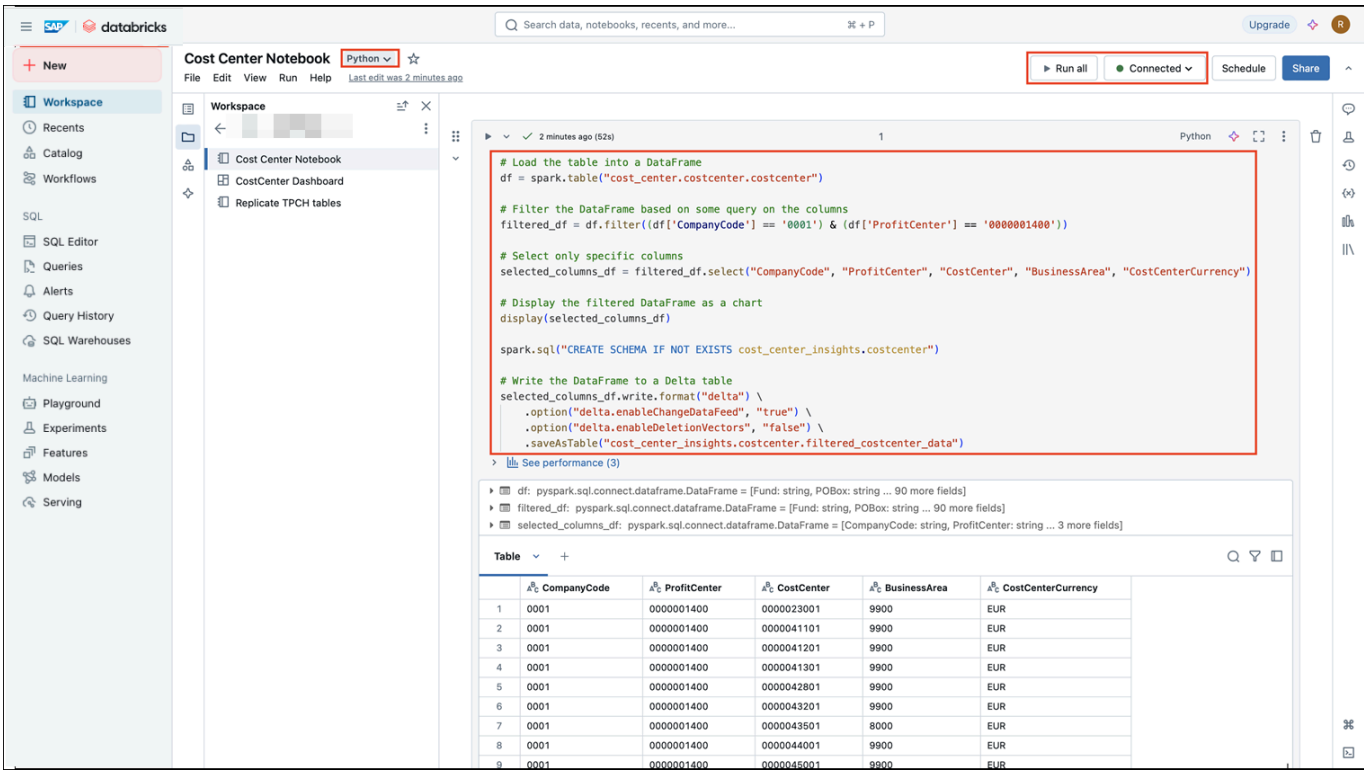
Relevant documentation: Get started: Query and visualize data from a notebook .

1. Create a new notebook by choosing the **(+) New** button in the sidebar in the top left corner in your SAP Databricks workspace.

2. Select the notebook language on the dropdown menu on the right side of the notebook title.

3. Select the **Serverless** compute on the right side of the **Run All** button on the top right corner.

4. Enter you query in one of the notebook cells and run (choose **Run all**, for example).



Querying Data in the SQL Editor

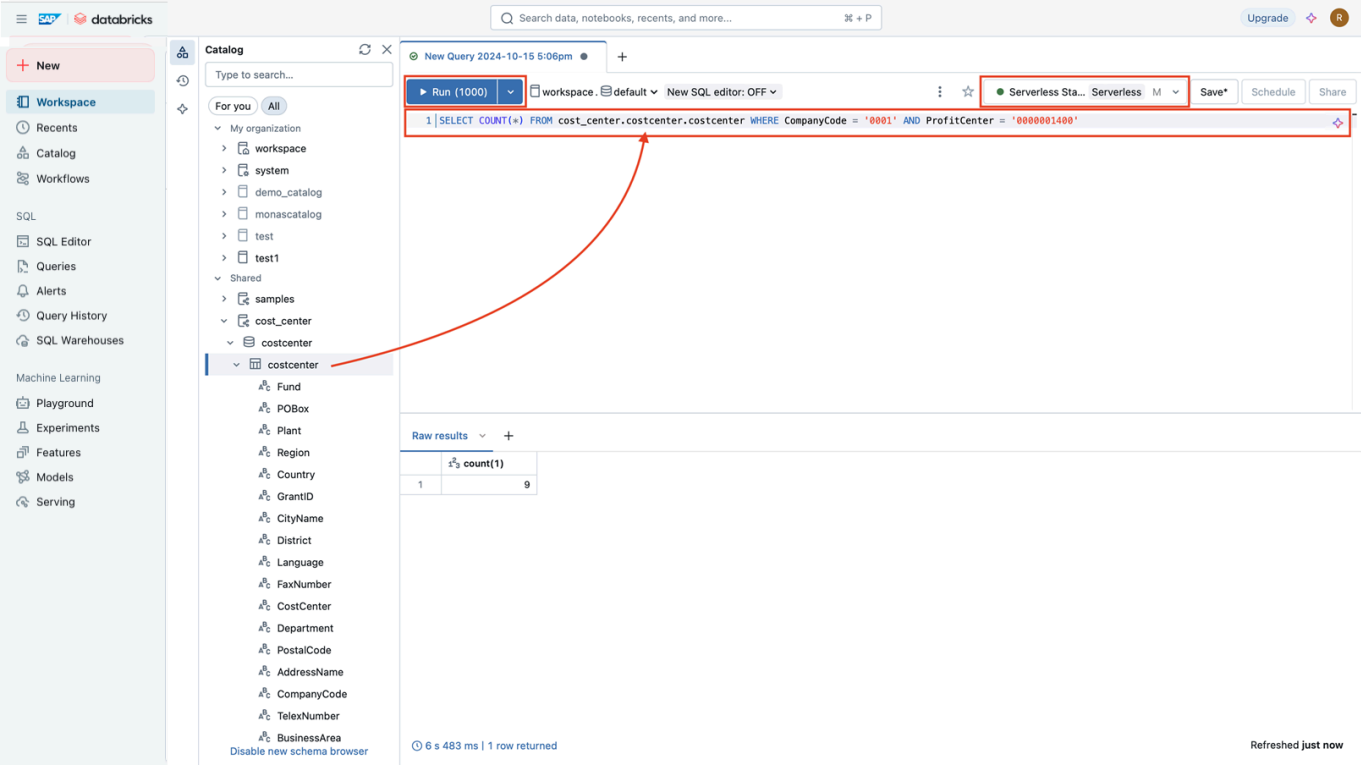
Relevant documentation: Write queries and explore data in the SQL editor .

1. Access the SQL Editor by choosing the **SQL Editor** button in the sidebar of the SAP Databricks workspace.

2. **Serverless SQL Warehouse** as the compute on the top right corner.

3. Type your query in the main text area.

4. Choose **Run**.



Creating and Working with Derived Data Products in SAP Databricks

The Unity Catalog integration with SAP Business Data Cloud enables bidirectional data flow by storing derived insights as Delta tables with rich ORD and CSN metadata. You can use this to expose SAP Databricks analytics results back to SAP applications like Datasphere for enterprise-wide consumption.

Prerequisites

You have added the necessary privileges. See Adding Privileges into SAP Databricks.

Creating a Catalog in the SAP Databricks Unity Catalog for Writing Derived Data Products

SAP Databricks documentation: Create catalogs .

Permissions

To create a new catalog in the Unity Catalog of SAP Databricks, the user must be a Data Product Admin" and have the following privileges on the SAP Databricks Metastore: USE PROVIDER, CREATE CATALOG.

Procedure

1. In the Catalog Explorer, choose **+** > **Add a catalog**.
2. Type the desired name, choose **Type as Standard**, uncheck **Use default storage** for **Use default storage**, and choose **Create**.
3. Go to the **Permissions** tab after accessing the created catalog and grant (at least) the following privileges to the users that will be able to write to this catalog: **USE CATALOG**, **USE SCHEMA**, **SELECT**, **CREATE SCHEMA** and **CREATE TABLE**.

Writing Data in the Form of Delta Tables to the Unity Catalog

When you create new Delta tables to expose as data products shared via Delta Sharing, you can configure table properties such as deletion vectors and change data feed based on your needs. Change data feed is disabled by default and can be enabled if required.

Permissions

To write the data to a catalog in the Unity Catalog, the user must have the following privileges on this catalog (granted by the catalog owner): USE CATALOG, USE SCHEMA, SELECT, CREATE SCHEMA and CREATE TABLE.

Writing Delta Tables in Python

In Python, the following command can be used to save a DataFrame named `df` as a Delta Table named `cost_center_insights.costcenter.filtered_costcenter_data` with deletion vectors and change data feed enabled:

Sample Code

```
df.write.format("delta") \
    .option("delta.enableChangeDataFeed", "true") \
    .option("delta.enableDeletionVectors", "true") \
    .saveAsTable("cost_center_insights.costcenter.filtered_costcenter_data")
```

Writing Delta Tables in SQL

In SQL, the following command can be used to create a Delta Table named `cost_center_insights.costcenter.filtered_costcenter_data` with deletion vectors and change data feed enabled:

Sample Code

```
CREATE TABLE cost_center_insights.costcenter.filtered_costcenter_data
  TBLPROPERTIES (
    delta.enableChangeDataFeed = true,
    delta.enableDeletionVectors = true
  )
  AS SELECT * FROM cost_center.costcenter.costcenter WHERE CompanyCode = '0001' and ProfitCenter
```

For more information, see Deletion vectors in Databricks and Use Delta Lake change data feed on Databricks .

Share the Delta Tables via Delta Sharing

In this step, a Delta Share will be created from data to be exposed to SAP Business Data Cloud.

Permissions

The user must have the following privileges:

- On the SAP Databricks Metastore: CREATE SHARE, USE RECIPIENT.
- On the catalog containing the data from the data product to be published: SELECT.

Note

Users with the **Data Product Publisher** role already have these permissions.

Procedure

1. From the sidebar menu, choose **Workspace** and, from the **Catalog Explorer**, select the schema or table that you would like to share. Choose the vertical ellipses and select **Share via Delta Sharing**.

   A dialog box appears.

2. Select **Create a new share with the Table/Schema** > Provide a share name, select the **sap-business-data-cloud** Recipient and choose **Share**.

Sharing Data Products from SAP Databricks to SAP Business Data Cloud

This section explains how to use Python SDK for publishing data products in SAP Business Data Cloud so that it becomes discoverable by other consumer applications in SAP Business Data Cloud, for example by SAP Datasphere.

Prerequisites

You have added the necessary privileges. See Adding Privileges into SAP Databricks.

Create a Delta Share

For a Delta Share to be exposed for downstream processing in SAP Databricks, it needs to be published as a data product with rich metadata. This can be done using an iPython Notebook available in your SAP Databricks Workspace. For further details on how to describe the data product, see Appendix A.

1. Access your SAP Databricks Workspace environment.
2. From the sidebar menu, choose **Catalog**.
3. Choose the **Manage** (Gear icon) button and, from the dropdown menu, select **Delta sharing**.
4. Under **Share data**, provide the following information:
   a. **Create share**: In the **Share name** field, enter a share name and choose **Save and continue**. The **Description** field is optional.
   b. **Add data assets**: Select one or more schemas, which includes all the tables in the schema, or one or more individual tables. Choose **Save and continue**.
   c. **Add notebooks**: (Optional) Specify the **Storage location**. Choose **Save and continue**.
   d. **Add recipients**: In the **Recipient** field, from the dropdown menu, select **sap-business-data-cloud**. Choose **Share data**.

ℹ Note

You can share schemas, tables, and views, and generate CSNs for them. Materialized views aren't supported in SAP Databricks.

The Delta Share is now exposed to the workspace.

Publishing the Delta Share as Data Product to SAP Business Data Cloud

1. Create a folder in the workspace by choosing Workspace > Create > Folder. Enter a name for this new folder and choose Create.

2. Create a notebook, in the newly created folder, by selecting that folder and choosing Create > Notebook.

3. Update the serverless environment version by following these steps:

   a. On the sidebar menu, choose the Environment button.

   b. In the Environment version field, use the dropdown menu to choose 3.

   c. Choose Apply.

4. Hover over the area under the rectangle and use +Code to create a cell in the notebook to run the instructions.

5. In this cell, install sap-bdc-connect-sdk:

   a. Execute pip install sap-bdc-connect-sdk to install the python SDK.

   b. Verify the SDK installation by executing pip show sap-bdc-connect-sdk. A successful output should look like this:

Output Code

```
Name: sap-bdc-connect-sdk
Version: 1.0.9
Summary: Python SDK for SAP Business Data Cloud
Home-page: https://www.sap.com
Author: SAP SE
Author-email:
License: SAP DEVELOPER LICENSE AGREEMENT
Location: /local_disk0/.ephemeral_nfs/envs/pythonEnv-738b110e-411e-47f8-bf8e-355d6c2b
Requires: argparse, cryptography, grpcio-status, protobuf, pydantic, python-dateutil,
Required-by:
```

6. Create or update the share in sap-bdc-connect-sdk.

   A share is a mechanism for distributing and accessing data across different systems. Creating or updating a share involves including specific attributes, such as @openResourceDiscoveryV1, in the request body, aligning with the Open Resource Discovery protocol. This procedure ensures that the share is properly structured and described according to specified standards, facilitating effective data sharing and management.

   Copy the below code snippet and paste it to the cell, then provide the share-name, title, shortDescription and description of your share.

Note

Ensure that the shortDescription and description are not the same.

Sample Code

```
from bdc_connect_sdk.auth import BdcConnectClient
from bdc_connect_sdk.auth import DatabricksClient

bdc_connect_client = BdcConnectClient(DatabricksClient(dbutils, "<recipient-name>"))

share_name = "<share-name>"

open_resource_discovery_information = {
    "@openResourceDiscoveryV1": {
        "title": "<title>",
        "shortDescription": "<short-description>",
        "description": "<description>"
    }
}
```

```
bdc_connect_client.create_or_update_share(
    share_name,
    open_resource_discovery_information
)
```

→ Tip

To get the <recipient-name>, go to Databricks Workspace > Catalog > Manage (Gear icon) > Delta Sharing > Shared by me > Recipients. The recipient name is under the field Name.

7. Create or update share CSN.

The CSN serves as a standardized format for configuring and describing shares within a network. To create or update the CSN for a share, it's advised to prepare the CSN content in a separate file and include this content in the request body. This approach ensures accuracy and compliance with the CSN interoperability specifications, facilitating consistent and effective share configuration across systems.

Copy the below code snippet and paste it to the cell, then provide the share-name of your share.

i Note

The CSN schema can be efficiently generated using a dedicated function generate_csn_template() available the in csn_generator file. This script automates the creation of CSN content for Databricks environments based on a Databricks share.

Sample Code

```
from bdc_connect_sdk.auth import BdcConnectClient
from bdc_connect_sdk.auth import DatabricksClient
from bdc_connect_sdk.utils import csn_generator

bdc_connect_client = BdcConnectClient(DatabricksClient(dbutils, "<recipient-name>"))

share_name = "<share-name>"

csn_schema = csn_generator.generate_csn_template(share_name)

bdc_connect_client.create_or_update_share_csn(
    share_name,
    csn_schema
)
```

→ Tip

To get the <recipient-name>, go to Databricks Workspace > Catalog > Manage (Gear icon) > Delta Sharing > Shared by me > Recipients. The recipient name is under the field Name.

If the generated CSN needs to be modified for any reason, it can be written to a file for editing, for example.

Sample Code

```
from bdc_connect_sdk.utils import csn_generator
import json

share_name = "<share-name>"
csn_schema = csn_generator.generate_csn_template(share_name)

file_path = f"csn_{share_name}.json"
with open(file_path, "w") as csn_file:
    csn_file.write(json.dumps(csn_schema, indent=2))

# change the file
```

```
with open(file_path, "r") as csn_file:
    changed_csn_schema = csn_file.read()

csn_schema = changed_csn_schema
```

8. Publish the data product.

   A Data Product is an abstraction that represents a type of data or data set within a system, facilitating easier management and sharing across different platforms. It bundles resources or API endpoints to enable efficient data access and utilization by integrated systems. Publishing a Data Product allows these systems to access and consume the data, ensuring seamless communication and resource sharing.

   Copy the below code snippet and paste it to the cell, then provide the **share-name** of your share. This will publish the share as a Data Product for consumption by other SAP services in Business Data Cloud.

   Sample Code

```
from bdc_connect_sdk.auth import BdcConnectClient
from bdc_connect_sdk.auth import DatabricksClient

bdc_connect_client = BdcConnectClient(DatabricksClient(dbutils, "<recipient-name>"))

share_name = "<share-name>"

bdc_connect_client.publish_data_product(
    share_name
)
```

   Tip

   To get the <recipient-name>, go to **Databricks Workspace** > **Catalog** > **Manage (Gear icon)** > **Delta Sharing** > **Shared by me** > **Recipients** >. The recipient name is under the field **Name**.

   Note

   The published data product will not appear instantly in SAP Datasphere, since it triggers a list of actions that take place in pre-scheduled intervals. If you don't see your data products, wait for a few minutes and try again.

Unpublishing a Data Product

Unpublishing a Data Product involves withdrawing access to the data, typically when it has been sufficiently consumed or is no longer desired to be shared. This action helps maintain control over data accessibility and ensures that sharing aligns with security and usage policies. By unpublishing, the system can effectively manage the lifecycle of data and its availability to other integrated platforms.

Copy the below code snippet and paste it to the cell, then provide the **share-name** of your share.

Sample Code

```
from bdc_connect_sdk.auth import BdcConnectClient
from bdc_connect_sdk.auth import DatabricksClient

bdc_connect_client = BdcConnectClient(DatabricksClient(dbutils, "<recipient-name>"))

share_name = "<share-name>"

bdc_connect_client.delete_share(
```

```
        share_name
    )
```

→ Tip

To get the <recipient-name>, go to Databricks Workspace > Catalog > Manage (Gear icon) > Delta Sharing > Shared by me > Recipients. The recipient name is under the field Name.

Next Steps

Once a data product has been published from SAP Databricks to SAP Business Data Cloud, you can use the catalog Data Product collection to view data products for use in your modeling and other projects. For more information, see Installing Data Products.

Appendix A. Describing the data product with ORD and CSN Metadata

A data product needs to be described in ORD (Open Resource Discovery) and CSN (Core Schema Notation), compliant to CSN Interop Specification v1.0 https://sap.github.io/csn-interop-specification/ .

The ORD information allows you to describe the content of the data product using text, links, tags, labels, etc. The CSN definition intends to describe the schema of the data, specifying the tables being shared, and their columns with the definition of data types, primary keys, associations/references, etc.

ORD Data Product definition

The ORD data product schema contains the complete list of attributes that compose the ORD data product metadata. However, not all fields can be set by users. For more details, check the following links:

- Fields that the user MUST provide:

  These are the fields that are mandatory in the data product object and cannot be automatically generated by the provider system. Therefore, they must be provided by the user publishing the data product.

Field	Mandatory	Example Value
title	yes	"Customer order"
shortDescription	yes	"Offering access to all online and offline orders submitted by customers."
description	yes	"The data product Customer Order offers access to all online and offline orders submitted by customers. It provides a customer view on the orders. For fulfillment-specific aspects please refer to the data product Fulfillment Order."

- Fields that the user can provide:

  These are the fields that are optional in the data product object and will not be automatically generated by the provider system. By default, they will not be present in the resulting data product unless defined by the user.

Field	Mandatory	Example value	Comments
labels	false	Sample Code { "label-key-1": [	Use to indicate reserved label keys that cannot

Field	Mandatory	Example value	Comments

		`"label-value-1",`
		`"label-value-2"`
		`]`
		`}`
			be overriden by the user.

deprecationDate	false	Sample Code `"2025-12-19T15:47:04+00:00"`	
sunsetDate	false	Sample Code `"2026-12-19T15:47:04+00:00"`	
successors	false	Sample Code `[ "customer.ext:dataProduct:other:v1" ]`	
industry	false	Sample Code `[ "Retail", "Public Sector" ]`	
lineOfBusiness	false	Sample Code `[ "Sales" ]`	
tags	false	Sample Code `[ "storage", "high-availability" ]`	
correlationIds	false	Sample Code `[ "sap.s4:communicationScenario:SAP_COM_0008" ]`	

dataProductLinks	false	Sample Code
		`[`
		`{`
		`"type": "support",`
		`"url": "https://example.com/"`
		`}`
		`]`	

links	false	Sample Code	

Field	Mandatory	Example value	Comments
		`[ { "title": "Example Link", "url": "https://example.com/", "description": "My example link description" } ]`	
documentationLabels	false	Sample Code `{ "My Documentation Label": [ "with link" ] }`	

- Fields that can be overwritten by the user:

Field	Mandatory	Example value	Comments
visibility	yes	"public"	Default value: "public". Alternative values: "interval", "private".
releaseStatus	yes	"active"	Default value: "active". Alternative values: "beta", "deprecated".

The minimum required fields that need to be specified are title , shortDescription and description .

Note

The **description** value must not contain the **shortDescription** value.

You can describe the ORD attributes for your data product on a JSON document. This JSON document needs to be created or uploaded to your SAP Databricks workspace (see Workspace files basic usage).

See an example of how the ORD information JSON document should look like:

Sample Code

```
{
  "title": "TPC-H",
  "shortDescription": "TPC-H is a decision support benchmark.",
  "description": "TPC-H consists of a suite of business-oriented ad hoc queries and concurrent d
}
```

CSN Interop Definition

In addition to the ORD metadata, the CSN Interop metadata is also required when publishing a data product. For a complete documentation of the CSN Interop specification, check CSN Interop Effective v1 • Interface Documentation . For examples of CSN documents compliant to this specification, check CSN Interop Effective v1 • Examples .

Requesting Help from SAP Technical Support

You can count on SAP Technical Support to help you resolve SAP Business Data Cloud issues when they arise. Following this procedure makes it easier for support engineers to understand your case and provide the right solution faster.

Prerequisites

Before creating a case, make sure you have:

- Checked the availability by service and region in the platform status. See SAP Trust Center.
- Reviewed the guidelines on how to engage with support. See Customer Support Best Practices and Offerings.
- Checked what to provide in an SAP BDC case and created a support user with the appropriate roles to allow an SAP support engineer to log in and troubleshoot your system. See SAP Note 3568017.

Procedure

To report a case to SAP Technical Support using SAP for Me, follow these steps:

1. Open SAP for Me and log in.
2. Open the **Services & Support** dashboard.
3. (Optional) You can use the **Knowledge Search** tab to find existing solutions in the following resources:
    - SAP Notes & KBAs
    - SAP Community
    - SAP Support Portal
    - Product Documentation
4. Use the **Get Support (Case, Expert Chat...)** card within SAP for Me to get assistance via the available support channels.

    For more information on how to report technical issues, see Get Support Application.

    You can receive SAP BDC service status notifications. See Manage Cloud Service Availability Notifications.

    For non-technical assistance, see Customer Interaction Center.

Related Information

SAP Business Data Cloud Community
Latest News on SAP Business Data Cloud